

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

1. All questions are compulsory.
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|---|---|
| A) Write down about Dapsone. | 2 |
| B) Write down structure and use of Quinine sulphate. | 2 |
| C) Outline the synthesis of Para amino salycilic acid. | 2 |
| D) Define anti-tubercular agents? Name the causative organism for tuberculosis. | 2 |
| E) Write down structure of Clindamycine and write down its uses . | 2 |
| F) Write down any two structures from biguanides. | 2 |
| G) Enlist the parameters of QSAR studies. | 2 |
| H) What is MOA of macrolide antibiotics. | 2 |
| I) What is Hammett's electronic parameter? | 2 |
| J) Give examples from sulphonamides. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|--|----|
| A) Classify Antiviral agent with example and structure. Give the mechanism of action, SAR, synthesis and uses of Acyclovir. | 10 |
| B) Discuss in detail chemical classification of antibiotics with example and structure. Discuss the SAR and MOA of tetracycline. | 10 |
| C) What are Antimalarial drugs? Discuss the etiology of malaria. Write down SAR and MOA of quinoline derivatives. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|---|---|
| A) Write a note on QSAR and its significance. . | 5 |
| B) Discuss about folate reductase inhibitors. . | 5 |
| C) Write down about quinolone as a urinary tract anti- infectives agents. . | 5 |
| D) Discuss in details about anthelmintics. | 5 |
| E) Write a short note on anti- protozoal agents. . | 5 |
| F) Write a note on prodrug concept and combinational chemistry in drug discovery. . | 5 |
| G) Write down short note on antifungal agent. | 5 |
| H) Outline the synthesis of Isoniazid and Metronidazole. . | 5 |
| I) Give a short note on beta lactamase inhibitors. . | 5 |

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Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- A) Define bacteriostatic and bactericidal agents with example.
- B) What is DOTS therapy?
- C) What are Immunosuppressants and immunostimulants? Give examples.
- D) What is chronopharmacology? Write its significance.
- E) What are mast cell stabilizers? Give its example
- F) What are the antacid? Give examples of non systemic antacids
- G) Explain the terms i. Teratogenicity ii. Mutagenicity
- H) Discuss the term biosimilars
- I) Name the antidotes of i. Lead poisoning ii. Arsenic poisoning
- J) Classify Semi-synthetic Penicillins

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- A) Differentiate between ampicillin and amoxycillin. Discuss pharmacology of Ampicillin in detail.
- B) What is peptic ulcer? Give pharmacological account of PPIs. Discuss multiple antibiotic approach in treatment of H. pylori infection.
- C) Discuss life cycle of malarial parasite with Classification antimalarial drugs. Give pharmacology of chloroquine.

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- A) Give chemical classification of antimicrobials. Enlist toxicities of tetracyclines and chloramphenicol.
- B) Explain general principles of treatment of poisoning.
- C) Classify anticancer agents with example. Give mechanism of action of alkylating agents.
- D) Classify Quinolones and Give mechanism of action of fluoroquinolones.
- E) Classify Sulphonamides. Give MOA of Co-trimoxazole.
- F) Discuss in detail pharmacology of acyclovir.
- G) Write note on urinary antiseptics.
- H) Define Immunosuppressants. Give brief account of cell mediated immunity and humoral mediated immunity and explain pharmacology of Cyclosporine in brief
- I) What are Antiamoebic drugs? Discuss pharmacology of metronidazole.

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Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|---|---|
| A) Write the advantages of biopesticides. | 2 |
| B) Define herb and herbal medicine. | 2 |
| C) Write the biological source, chemical constituents, and uses of Ginseng. | 2 |
| D) Write the role of natural flavours and perfumes. | 2 |
| E) Comment on Farmer's right and Biopiracy | 2 |
| F) Write in short about viscosity builders and disintegrants with a suitable example. | 2 |
| Define IPR and Patent. | 2 |
| What is the composition of Vermipost? | 2 |
| Define organic farming and mention the uses of organic fertilizer. | 2 |
| Write scope of nutraceuticals. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|---|----|
| A) Discuss the Indian system of medicines. Enumerate the basic concepts, procedures, diagnosis, and treatment behind Ayurveda. | 10 |
| B) Explain in detail about WHO guidelines for the quality control of herbal drugs. | 10 |
| C) What are herb-food and herb-drug interactions, and how do they support the answer with a suitable example of Hypericum and Garlic? | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|---|---|
| A) Give an account of Herbs used in the Skin cosmetic preparations. • | 5 |
| B) Explain in brief good agriculture practices in medicinal plants. • | 5 |
| Write a detailed note on Schedule-T. Good manufacturing processes of the Indian system of medicine. • | 5 |
| C) Write about pest and pest management. • | 5 |
| E) Give a brief account of plant-based industries and institutions in India. | 5 |
| F) Write a note on preparation and standardization of Asava and Arishta. • | 5 |
| G) Write the health benefits of Fenugreek and Garlic. • | 5 |
| H) Write in detail the patent case study of any one Curcuma or Neem. • | 5 |
| I) Write composition of ASU-DTAB, and ASU-DCC and their responsibilities. | 5 |

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Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|---|---|
| A) What is mean by Passive Diffusion? | 2 |
| B) Give the classification of Pharmacokinetic Models. | 2 |
| C) Enlist various theories of Dissolution. | 2 |
| D) What are the various mechanisms involved in absorption of drugs from GI Tract? | 2 |
| E) Write BCS Classification of drug with examples. | 2 |
| F) List out various dissolution test apparatus as per USP. | 2 |
| G) What is Loading Dose and Maintenance Dose? | 2 |
| H) What is Apparent volume of distribution? Mention factors affecting on it. | 2 |
| I) Define Bioavailability. Give factors which affect bioavailability of drug. | 2 |
| J) What is Total Clearance? | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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|---|----|
| A) Discuss in detail factors affecting protein binding of drugs. Give the significance of protein/tissue binding of drug. | 10 |
| B) Explain one compartment open model for assessment of parameter by IV Infusion administration. | 10 |
| C) What is Non-linear Pharmacokinetics? Enlist various reasons of non-linearity along with the significance of non-linear pharmacokinetics. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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|--|---|
| A) How pharmaceutical factors affect the absorption of drugs? • | 5 |
| B) Explain Plasma drug concentration-time profile along with its pharmacokinetic parameters. • | 5 |
| C) Explain Michaelis-menton method of estimating parameters. • | 5 |
| D) Write a note on In-Vitro-In-Vivo Correlation. • | 5 |
| E) Define Area Under the Curve. And describe various methods to determine AUC. | 5 |
| F) Describe in detail methods of enhancement of bioavailability of drug. • | 5 |
| G) Explain one compartment open model for assessment of pharmacokinetic parameters related to plasma data for IV Bolus administration. • | 5 |
| H) Write a note on multi-compartment models. • | 5 |
| I) Explain non-renal excretion of drugs. | 5 |

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Instructions to the Students:

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Draw diagrams / figures wherever necessary

Figures to right indicate full marks

Objective Type Questions (Answer all the questions) (10 x 2) = 20

- What is role of primer in PCR.
- Write the functions of MHC.
- Define terms- A) Transformation B) ELISA
- Enlist the enzymes used in gene cloning.
- What is Biotransformation?
- Define vector. Write the features of good vector
- Write Advantages and Disadvantage of solid-state fermentation
- Which enzyme is used to cut DNA duplex?
- Name four mutagenic agents
- Enlist the applications of recombinant DNA technology

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Long Answers (Answer 2 out of 3) (2 x 10) = 20

- Elucidate the term PCR and Explain in brief with diagram.
- Explain different types of biosensor & application of biosensor used in diagnostics.
- Describe in detail different biotransformation reaction with special reference to steroids.

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Short Answers (Answer 7 out of 9) (7 x 5) = 35

- What is vaccine? & explain the types of vaccines.
- Describe in brief different types of fermenters.
- What is basis of type-II hypersensitivity reaction? Explain with suitable example.
- Explain the process of ELISA test & its application.
- Explain the general methods of recombinant DNA technology.
- Describe in brief genetic organization of eukaryotes.
- What is enzyme immobilization? Explain the methods of enzyme immobilization.
- What is mutation? Explain in details the types of Mutation.
- Explain the principle of protein engineering with neat labelled flow chart.

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Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- A) Secondary Packaging materials.
- B) Batch formula records.
- C) Standard operating procedure.
- D) Roles of head production department.
- E) Waste disposal.
- F) What is warehousing & material management.
- G) Define qualification & calibration.
- H) Classify packaging material used in pharmaceutical industry with example.
- I) What is contamination and cross contamination?
- J) Enlist Q series of ICH guideline.

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- A) What is Validation? Explain types of validation & Discuss Validation master plan.
- B) Write a note on Design, Construction, Layout & Maintenance.
- C) Discuss the importance of GLP? Explain protocol content of Non-clinical laboratory study.

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- A) Explain the various steps involve in complaint handling.
- B) Discuss the qualification of UV-Visible Spectrophotometer.
- C) Compare QA & QC as per GMP.
- D) Write in detail on ISO 9000 & ISO 14000.
- E) What is Quality by Design? Note down its tools briefly.
- F) Discuss the quality control test for rubber closures.
- G) Write a note on NABL.
- H) Explain about personnel responsibilities, training & hygiene.
- I) Discuss the QSEM guideline as per ICH.